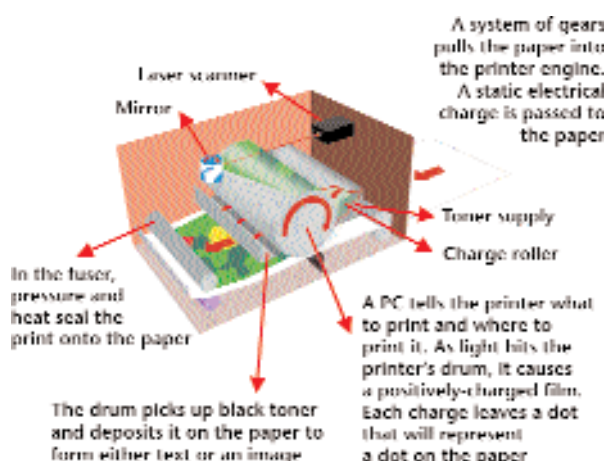


## Under the Bonnet

Before a laser printer can start printing, it needs to have all the information about a page in its memory. Where information about the image to be printed is communicated to it via a page description language (PDL), the printer's internal processor first converts the instructions into a bitmap, which is stored in the printer's memory, to be later reproduced on paper.

At the heart of the printer is a small rotating drum—the organic photo-conducting cartridge (OPC)—with a coating that allows it to hold an electrostatic charge. First the drum is given a positive charge and then a laser beam scans across the surface of the drum, selectively imparting points of negative charge onto the drum's surface that represents the output image—every point on the drum corresponds to a point on the sheet of paper. In the meantime, the paper is passed through an electrically charged wire that deposits a negative charge onto it. The laser beam is then turned on and off as it scans the rotating drum, using a complex arrangement of spinning mirrors and lenses. The mirror drum spins incredibly fast and is synchronised with this on/off switching of the laser. A typical laser printer will perform millions of switches every second. Inside the printer, the drum rotates to build one horizontal line at a time—the smaller the rotation, the higher the resolution down the page (the step rotation on a modern laser printer is typically 1/600th of an inch, giving a 600 dpi vertical resolution rating and in some of the latest printers, this resolution even goes up to 1440 dpi. Similarly, the faster the laser beam is switched on and off, the higher the resolution across the page. As the drum rotates to present the next area for laser treatment, the written on area moves into the laser toner.

Toner used for printing mono is very fine black powder, positively charged so that it is attracted to the points of negative charge on the drum surface. In colour printing, it could be either a CMYK or RGB used individually or in a combination. Thus, after a full rotation, the drum's surface contains the whole of the required black image

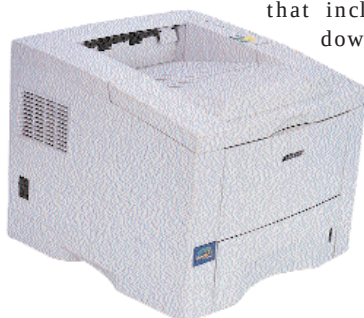


for mono printing or any varied settings for colour.

A sheet of paper now comes into contact with the drum, fed in by a set of rubber rollers. The charge on the paper is stronger than the negative charge of the electrostatic image, so the paper magnetically attracts the toner powder. As it completes its rotation, it lifts the toner from the drum, thereby transferring the image to the paper. Positively charged areas of the drum don't attract toner and result in white areas on the paper. Toner is specially designed to melt very quickly and a fusing system now applies heat and pressure to the imaged paper in order to stick the toner permanently. The drum is then cleaned of any remnants of toner either by mechanically scraping the drum or by covering the drum with an even electrical charge so the laser can write on it again.

For managing the printer for a longer run, the felt pad which cleans the drum and the corona wire, needs to be changed regularly.

**Resolution:** While buying a printer you should first define your needs. If you need a device for printing documents at the maximum possible quality, you should go in for a printer that supports the maximum resolution (1200 dpi and above). A 1200 dpi printer will give you the capability to print high-quality monochrome images or complex documents that include many details. The only



*With its high resolution of 1200 dpi, the Samsung ML-1450 is a good choice for crisp and clear printouts*

downside is that these high-resolution printers are anywhere between 25 per cent and 60 per cent more expensive than their 600 dpi counterparts. You can opt for a printer with a lower resolution of say, 300 dpi, if you don't intend to print documents at extremely high quality and if you need a cost-effective solution. Out of the 10 printers that we received, six supported a res-

olution of 1200 dpi, the remaining ones supported a resolution of 600 dpi. The six high-resolution printers were Canon LBP 1120, Epson EPL-5900L, HP Laserjet 1200, Lexmark E322, Samsung ML-1450 and Wipro Laser 1540.

**Interface:** Most of the printers connect to the PC through the popular parallel interface. But now manufacturers have also started to produce printers with the USB interface and some even provide a USB and parallel interface in combination. If you own an older machine that does not feature a USB interface, a printer which has only a parallel interface will suffice, but if you own a newer computer that features a USB port as well as parallel ports, a USB printer is preferable. All the printers in the comparison sported a USB port, but the HP Laser 1000 used a USB-to-parallel port interface and, integrating the old with the new, the HP Laserjet 1200 featured the USB 2.0 interface in addition to a parallel port. **Paper capacity:** One must also consider the paper capacity of the paper-in tray. The Samsung ML-1450 offered a mammoth paper capacity of 550 sheets. This printer does not support the option for manual feed, but that is hardly of any concern while buying laser printers. This one is a great option if your print jobs are large in quantity, where you would not want to be bothered with having to fill the paper tray too often.

The next highest tray capacity in this comparison was of 250 sheets. The printers that offered this included the Canon LBP 1210, the HP Laserjet 1200, the Laserjet 1000, and the Wipro Laser 1540. The printer that supported the lowest tray capacity was the Canon LBP 1120 with support for 125 sheets—however, this was also among the lowest priced printers in this roundup.